INTERNAL TECHNICAL BRIEFING

Voice AI Platform — Core Deployment Challenges

Solution Design, Cost–Efficiency Trade-offs and Implementation Sequence

Field	Detail
Document Type	Technical solution briefing for management review
Subject	Eight core challenges in deploying voice AI agents for Indian real estate presales, with proposed solutions
Market Focus	NCR (National Capital Region), Indian real estate developers and channel partners
Prepared For	Engineering / Product Management review
Design Constraint	Every solution must be effective at solving the stated problem while remaining defensible on cost
Status	For review and prioritisation

Purpose of this document: To move the discussion from "here are the risks" to "here is what we build, in what order, and what it costs us." Each challenge below is stated briefly, then answered with a concrete engineering or commercial solution.

1. Introduction

Demand for voice AI in India is not the constraint. The Indian voice AI market was valued at approximately $153M in 2024 and is projected to reach roughly $958M by 2030 at a 35.7% CAGR, with a large share of top-tier banks already running production voice agents. The constraint is deployment reliability — the rate at which pilots fail to convert into contracts.

That distinction matters for how we allocate engineering effort. If demand were the problem, we would invest in market development. Because reliability is the problem, we invest in the operational layers that determine whether a pilot survives: data quality, latency on Indian cellular networks, answer accuracy, graceful failure, regulatory posture, and organisational adoption inside the customer.

This document sets out eight challenges that determine that outcome, and for each one a solution designed against a hard commercial constraint: the fix must be effective, and it must not inflate our per-minute cost of delivery beyond what the market will pay.

1.1 Scope

- **In scope:** outbound and inbound presales voice agents for residential real estate; lead qualification, site-visit scheduling, follow-up and inbound concierge flows.
- **In scope:** the technical architecture, unit economics and commercial packaging decisions that follow from each challenge.
- **Out of scope:** vendor selection and final pricing negotiation; detailed vendor rate cards are to be confirmed separately before financial modelling.
- **Out of scope:** post-sale customer service and collections flows, which sit in a later phase.

1.2 How to Read This Document

Section 2 establishes the cost model and the architectural principles that most of the solutions depend on — reading it first makes the individual solutions shorter to follow. Sections 3 through 10 address one challenge each, in a consistent format: the challenge, why it matters, the proposed solution, key implementation points, and a verdict. Sections 11 onward consolidate everything into summary tables, an implementation sequence, and a risk register.

2. Cost Model and Architectural Foundations

Before addressing individual challenges, it is worth stating plainly where our money goes, because that determines which solutions are affordable.

2.1 Where Cost Accumulates

Our marginal cost is measured per minute of call, and it accumulates across four components:

- **Telephony:** SIP trunk and per-minute carrier charges. Largely fixed by the carrier; the only lever is calling fewer bad numbers.
- **Speech-to-text (STT):** charged per minute of audio processed. Scales linearly with call duration.
- **LLM inference:** charged per token. This is the component with the widest variance and therefore the largest optimisation opportunity.
- **Text-to-speech (TTS):** charged per character or per second of audio generated. Highly compressible through caching.

The important observation is that LLM inference and TTS — the two most controllable components — are also the two where naive implementations waste the most. A system that sends every conversational turn to a frontier model and synthesises every sentence fresh is paying full price for work that is, in a presales script, highly repetitive.

2.2 The Governing Principle: Tier Everything

The single most effective cost control available to us is not a cheaper vendor. It is routing each unit of work to the cheapest tier capable of handling it. Applied to conversational turns, this produces a three-tier router:

Tier	What It Handles	Approx. Share of Turns	Marginal Cost
T0 — Deterministic state machine (no model call)	Greeting, AI disclosure, consent capture, opt-out handling, slot confirmation, closing lines	High	Effectively zero
T1 — Small / fast model	Slot extraction, structured FAQ answers, intent classification, simple confirmations	Moderate	Low
T2 — Frontier model	Objection handling, confused or off-script callers, NRI and financing edge cases, negotiation-adjacent turns	Low	High

Assumption to validate: the tier distribution above is our working hypothesis based on the shape of a presales script, not a measured result. It must be verified against a real call sample before we commit to a price point. If T2 turns out to carry a materially larger share of turns, our per-minute cost assumption changes and the commercial model needs revisiting.

Routing is decided on intent confidence and turn type. Architecturally this fits our existing modular monolith cleanly — tier selection is a field on the configuration profile, not a new service, so it adds no operational surface area.

2.3 Four Supporting Mechanisms

Four further mechanisms recur across the solutions in this document. They are introduced once here and referenced later.

(a) Pre-rendered TTS cache

A large share of what the agent says in a presales call is fixed text: the disclosure line, the project name, slot proposals, brochure handoff, closing lines. These are rendered to audio once per project, stored, and spliced with streaming TTS only for the genuinely dynamic spans. This reduces TTS spend and drives time-to-first-audio on those turns close to zero. The trade-off is audible seams if splicing is done carelessly — splices must occur at sentence boundaries, with identical voice settings and a short crossfade, and never mid-sentence.

(b) Structured facts separated from prose

Anything numeric or enumerable — price by unit type and floor, possession date, RERA number, payment plan terms, current availability — is held in a structured store and retrieved by deterministic lookup. Only narrative content is served through retrieval-augmented generation. This is the difference between an agent that is confidently wrong about a price and one that is either correct or explicitly defers.

(c) Pre-dial data quality gate

A deterministic validation stage that every lead must pass before a call is placed. It involves no model inference, costs effectively nothing to run, and prevents the failure mode most likely to end a pilot early. Detailed under Challenge 3.

(d) Latency budget as line items

Latency is managed as a set of individually attributable components with individually attributable fixes, not as a single number to be optimised by intuition. Detailed under Challenge 2.

3. The Eight Core Challenges

Each of the following sections addresses one challenge. The header table on each gives a fast read on severity and cost; the verdict box at the end states the position we are taking.

Challenge 1 — The Uncanny Valley Problem

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
High	Conversation design	Low	Neutral — no added inference

The Challenge

Callers disconnect on AI voice calls, but the reason is more specific than a general preference for humans. Voices trained toward broadcast-quality delivery — perfect enunciation, flawless grammar, no hesitation — measurably reduce trust rather than building it. A founder who tested AI calling described the first batch as a disaster precisely because the voice was too clear and too perfect; recipients identified it as synthetic immediately.

The failure is compounded when the agent cannot switch naturally between Hindi and English within a sentence, which is how most NCR buyers actually speak. The moment the caller identifies the voice as artificial, their defences go up and the remainder of the call is spent managing suspicion rather than qualifying the lead.

Why It Matters

- This is a first-impression failure, so it caps the performance of everything downstream — a lead lost in the first ten seconds never reaches qualification, scheduling or follow-up.
- It is not fixable by buying a better voice model. Improving audio fidelity moves us further into the uncanny valley, not out of it.
- It is directly measurable through early-call hang-up rate, which makes it one of the few challenges here with a fast feedback loop.

Proposed Solution

Engineer imperfection deliberately. The objective is not the highest-quality voice available but the most natural one, which means introducing the specific irregularities that human speech contains and that synthetic speech lacks.

Key Implementation Points

- **Filler injection.** Insert conversational fillers ("hmm", "achha", "ek second") at the points where the system is waiting on a model response. The filler serves two purposes simultaneously — it humanises the delivery and it covers real processing latency at zero additional cost.
- **Micro-pauses and breath.** Short pauses at clause boundaries and an audible breath at sentence breaks. Both are configuration-level changes, not model changes.
- **Full barge-in support.** When the caller interrupts, TTS playback must stop within roughly 200ms. An agent that talks over an interrupting caller is identified as a machine faster than one with an imperfect voice.
- **Native code-switching.** Hindi–English mixing must be handled within a single utterance by a voice designed for Indian prosody, not by switching voices or languages mid-call.
- **Test on real network conditions.** All voice validation must run over actual Jio and Airtel cellular calls at 8kHz codec quality. WiFi and browser-based demos systematically overstate quality and will mislead us.

- **Tone adaptation.** Adjust pace and register based on detected caller state — a caller who sounds rushed should get a shorter script path.

> **Verdict:** Do not chase perfect voice quality; chase natural voice quality. This challenge is solved through conversation design and configuration, not through model spend. Effort is low and it adds nothing to our per-minute cost — filler audio is pre-rendered and served from cache.

Challenge 2 — Latency on Indian Networks

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
High	Infrastructure and pipeline design	Medium	Positive — reduces cost

The Challenge

The voice pipeline runs speech-to-text, then LLM inference, then text-to-speech, and each stage contributes delay. A total of around 800ms is acceptable on US broadband. On an Indian cellular connection with codec compression, that same 800ms becomes roughly 1.5 to 2 seconds — past the point at which natural conversation breaks down.

The consequence is not merely that the agent feels slow. Callers interpret silence as a dropped call, start speaking over the agent, or hang up. A demo with an excellent voice and a two-second gap after every sentence will fail in production regardless of how good the underlying model is. India is not a smaller version of the US market; it is a materially harder deployment environment.

Why It Matters

- Latency is the failure mode that is easiest to miss in development and impossible to miss in production, because development happens on good networks.
- It interacts with Challenge 1 — awkward silence is one of the strongest signals that the caller is speaking to a machine.
- Several of the fixes reduce cost as well as latency, so this work pays for itself rather than competing with the cost constraint.

Proposed Solution

Stop treating latency as a single number to be reduced and start treating it as a budget with named line items, each with its own owner and its own lever. Our target is sub-500ms end-to-end. The table below allocates that budget.

Pipeline Component	Why It Costs Time	Lever
Endpointing / voice activity detection	A fixed silence timeout waits the full duration on every turn, whether or not the caller has finished	Semantic turn detection instead of a fixed timer
STT transcription	Batch processing waits for the utterance to complete before starting	Streaming STT with partial transcripts
LLM time-to-first-token	Long system prompts and large retrieved context must be processed before generation begins	Prompt caching on the system prompt and knowledge block; retrieve top-2 chunks only
TTS time-to-first-byte	Waiting for the complete model response before synthesis begins	Begin synthesis at the first sentence boundary
Network transit	Cross-region hops between telephony, model and application layers	Co-locate everything in the Mumbai region; keep the media path domestic
Perceived delay	Any residual gap reads to the caller as a dropped call	Inject a filler on escalation to the frontier model — buys roughly 800ms of real budget

Key Implementation Points

- **Semantic endpointing.** This is usually the largest hidden contributor. A fixed silence timeout of 700ms adds 700ms to every single turn. Turn-detection based on semantic completeness rather than a fixed timer recovers most of that.
- **Streaming, not batch.** STT must emit partial transcripts continuously so downstream stages can begin work before the caller finishes speaking.
- **Prompt caching.** Cache the system prompt and the project knowledge block. This reduces time-to-first-token and cuts input token cost on every turn — a latency fix and a cost fix in the same change.
- **Retrieve narrowly.** Return the top two retrieved chunks, not the top ten. Retrieval breadth drives both prompt size and inference latency, and in practice adds little accuracy on well-structured project data.
- **Start TTS early.** Begin synthesis at the first sentence boundary rather than waiting for the full model completion.
- **Keep the media path in India.** Host in the Mumbai region and ensure audio does not leave the country mid-call. Cross-region hops are pure, avoidable delay.
- **Buy perceived time.** Where a turn genuinely requires the frontier model, play a filler while it runs. This buys real processing budget at no cost and is invisible to the caller.

> **Verdict:** Latency is an engineering priority, not a polish item, and it must be budgeted line by line. Note that prompt caching and narrow retrieval reduce cost while reducing latency — this challenge is one where the effective solution and the cheap solution are the same solution.

Challenge 3 — Dirty Data and CRM Chaos

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
Critical	Data quality	Low	Positive — reduces wasted minutes

The Challenge

This challenge receives less attention than voice quality and ends more deployments. AI agents are only as good as the records they are given. In NCR real estate the data condition is consistently poor: phone numbers are wrong, the same buyer appears three times from three portals, and lead status has not been updated in weeks.

The resulting failure is highly visible and highly damaging. Our agent calls a lead that was sold last month, or calls the same person three times in a day from three different source records. The developer does not conclude that their CRM is untidy — they conclude that our product is unreliable, and the pilot ends.

Why It Matters

- It converts an internal data problem into an external product-quality perception, which is the worst possible trade.
- It wastes call minutes directly — every dial to a dead or duplicate number is telephony and STT spend with zero possible return.
- It is entirely preventable with deterministic code, which makes it the highest return-on-effort item in this document.

Proposed Solution

Insert a mandatory data quality gate before the dialler. This runs as pure application logic with no model inference, so its marginal cost is effectively zero, and it must be a hard gate — no lead is dialled without a passing verdict.

Key Implementation Points

- **Normalise and validate.** Convert every number to E.164 format and validate against Indian numbering series before anything else runs.
- **Check reachability before dialling.** A network lookup on the number filters dead and disconnected lines before we spend call minutes on them. This single step recovers its own cost.
- **Deduplicate in cost order.** Run exact phone-hash matching first because it is free and catches the majority of duplicates. Apply fuzzy name and email matching only to the residual set.
- **Reconcile status with the CRM.** Pull current lead status immediately before dialling and maintain a suppression list covering sold, lost and recently-contacted records.
- **Scrub against DND.** Run the registry check on every batch and retain timestamped evidence of the check having been performed.
- **Log every verdict.** Write the gate decision and its reasoning to our decision trace store. When a developer disputes a call, we answer with a record rather than an apology.
- **Report quality back to the customer.** Surface the rejection statistics to the developer. It reframes their data problem as our diagnostic capability and creates a reason for them to fix it.

> **Verdict:** This is the lowest-cost and highest-return item in this document. It is unglamorous, requires no model spend, and prevents the single most common cause of pilot failure. It should be built first.

Challenge 4 — Tightening Regulation

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
High	Compliance and positioning	Low	Neutral

The Challenge

In 2025 TRAI strengthened anti-spam regulation — tighter rules for telemarketers, extended complaint windows, and greater oversight of automated dialling. Those measures verify the origin of a call but do not tell the consumer whether they are speaking to a person or a machine. India's emerging AI governance framework places growing weight on transparency, disclosure and informed consent where AI interacts directly with consumers.

The direction of travel is clear: mandatory disclosure that the caller is an AI system. If that becomes law, any product positioned on sounding indistinguishable from a human loses its core claim overnight.

Why It Matters

- It is a positioning risk before it is a compliance risk. A product whose value depends on concealment is one regulation away from being unsellable.
- Enterprise developers will begin asking about AI disclosure and data handling during procurement, ahead of any legal requirement. Being ready is a sales advantage.
- Retrofitting consent and audit trails into a live system is significantly more expensive than building them in now.

Proposed Solution

Build disclosure in from the first call and treat it as a trust-building feature rather than a compliance tax. An agent that announces itself and then demonstrates precise knowledge of the project generates more confidence than one that is caught pretending.

Key Implementation Points

- **Pre-rendered disclosure.** The opening disclosure line is fixed audio played deterministically at call start. It cannot be paraphrased away by a model, costs nothing at runtime, and is identical on every call — which makes it auditable.
- **Opt-out without model involvement.** Detect opt-out phrases by keyword spotting on partial STT transcripts. No LLM sits in this path, so the response is immediate and cannot be reasoned around. Termination and suppression-list write happen in the same step.
- **Maintain a consent ledger.** Record timestamp, DLT template reference, confirmation that disclosure played, opt-out status and retention window for every call.
- **Align with DPDP principles.** Purpose limitation, a defined retention period, and a working deletion-on-request path. These are procurement questions we will be asked.
- **Shift the value proposition.** We do not sell an agent that sounds human. We sell an agent that answers project questions accurately at 9pm on a Sunday. That claim is regulation-proof.

> **Verdict:** Roughly one day of engineering removes an entire category of regulatory exposure and converts a looming threat into a differentiator. Do it now rather than when it becomes mandatory.

Challenge 5 — The First Thirty Seconds

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
Critical	Knowledge architecture	Medium	Positive — structured lookup is cheaper than retrieval

The Challenge

This is the real-estate-specific form of the hang-up problem. Property is a high-value, high-emotion purchase, and buyers expect precise answers. An agent that stumbles on a basic question, gives a vague response, or states a wrong figure loses the lead permanently — and the buyer's negative impression attaches to the developer's brand, not just to us.

The concrete failure looks like this: a buyer asks "3BHK ka price kya hai 12th floor pe?" and receives an approximate or incorrect answer. That lead is gone, and the developer now has a reason to end the pilot. A generic answer to a specific question is worse than no call at all.

Why It Matters

- Accuracy on project specifics is the difference between an AI receptionist and an AI salesperson. It is our primary defensible advantage over generic voice platforms.
- The most damaging failure mode is confident incorrectness, which is exactly what unstructured retrieval over a brochure produces when asked for a number.
- Knowledge base quality is the input that most directly determines conversion, so it justifies proportionally more attention than voice tuning.

Proposed Solution

Split the knowledge base by data type and never allow retrieval to answer a numeric question. Facts that are enumerable are looked up deterministically; only narrative content passes through retrieval.

Key Implementation Points

- **Structured facts, deterministic lookup.** Price by unit type and floor, possession date, RERA number, payment plan terms, current availability and inventory status live in a structured store and are retrieved by exact query. There is no generation step and therefore no opportunity to be wrong.
- **Prose through retrieval.** Location advantages, amenity descriptions, connectivity narrative and market comparison are served through the retrieval pipeline, where approximate phrasing is acceptable.
- **Explicit deferral on gaps.** If a requested field is absent, the agent says so and offers a human callback. It never generates a plausible value. This rule is absolute.
- **Templated onboarding.** A single structured ingestion sheet per project, populated by a one-time model pass over the brochure and price list, then reviewed by a human before go-live. Onboarding cost is amortised across the project lifetime rather than paid on every call.
- **Version the knowledge base.** Prices and inventory change. Each project knowledge base carries a version and an effective date, and stale data triggers an alert rather than silently misinforming buyers.
- **Front-load the strongest content.** The first exchange should demonstrate specific project knowledge immediately, because that is the moment the caller decides whether the conversation is worth continuing.

> **Verdict:** This is our accuracy moat and it should be treated as such. It is also cheaper at runtime than the alternative — a structured lookup costs nothing per query, while retrieval consumes tokens on every turn. Correctness and cost point the same direction.

Challenge 6 — Multi-Turn Complexity and Edge Cases

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
High	Dialogue management	Medium	Slight increase — bounded by confidence gating

The Challenge

Real calls are disorderly. The buyer is driving, a child is crying in the background, and they move from asking about price to asking about parking to saying "ek minute hold karo." Most voice agents handle the intended path well and collapse on the first unexpected turn, falling back to a rigid decision tree that alienates the caller.

The genuine test of a dialogue system is what happens when the caller interrupts mid-utterance with an unrelated question. Rigid script-bound flows fail this test, and failing it in front of a buyer making a major financial decision is expensive.

Why It Matters

- Handling digression well is directly visible to the buyer and is the clearest differentiator against competitors running fixed flows.
- Poor handling produces loops, which are the most frustrating possible failure and the one most likely to be repeated back to us as evidence the product does not work.
- Edge-case behaviour cannot be validated by scripted testing, so it needs a different testing method than the rest of the system.

Proposed Solution

Give the dialogue manager a digression stack and a confidence gate, then validate both against real call transcripts rather than scripted flows.

Key Implementation Points

- **Digression stack.** When the caller changes topic mid-flow, push the current objective onto a stack, answer the digression, then return with a natural re-entry line ("waise, aap poochh rahe the..."). This is inexpensive to implement and is precisely what makes rigid competitors sound rigid.
- **Confidence gating.** Two consecutive low-confidence turns trigger a warm transfer to a human. Not a third attempt. Looping is a worse outcome than transferring, and this rule also caps the cost of a difficult call.
- **Explicit graceful fallback.** A defined handoff line — offering to connect the caller with the sales team — is always available and is preferred over improvisation when the agent is uncertain.
- **Hold and noise handling.** Recognise hold requests, tolerate background noise, and handle mid-utterance interruption without losing conversational state.
- **Build a golden set from real calls.** Record and transcribe fifty genuine presales calls from a developer's own team. Every unusual turn in those calls becomes a regression test case. This is the only reliable way to find edge cases before customers do.
- **Run regression in text mode.** Execute the golden set as text on every change, which is fast and inexpensive. Sample a subset for full audio evaluation rather than running audio on every iteration.

> **Verdict:** Test against real conversational mess, not against our own scripts. The digression stack is the visible differentiator; the confidence gate is what keeps a difficult call from becoming an expensive one.

Challenge 7 — The Adoption Gap — Buyer Is Not the User

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
High	Organisational change	Zero engineering	None

The Challenge

The person who buys our product — a VP of Sales or Marketing Head — is not the person who uses it. The presales team is. That team frequently perceives the AI as a threat to their role and undermines the pilot in ways that are difficult to attribute: CRM records go unupdated, AI-qualified leads are described as poor quality, and follow-up on AI-generated leads is slow enough that the conversion loop never closes.

This is an organisational change management problem wearing the costume of a technical problem. No amount of engineering fixes it, and misdiagnosing it as a product defect leads to wasted development effort.

Why It Matters

- A pilot can fail on adoption while the technology performs correctly, and the post-mortem will still blame the technology.
- The failure signals are indirect and easily mistaken for product issues, which makes them dangerous to our roadmap prioritisation.
- The fix costs nothing to implement, which makes ignoring it indefensible.

Proposed Solution

Design the pilot so that the AI cannot reasonably be perceived as taking work away from the team, and so that it visibly makes their existing work easier.

Key Implementation Points

- **Scope the pilot to unwanted work.** During the pilot, give the AI only the aged, dormant and after-hours lead pool — calls the team is not making anyway. There is no work to defend, so there is no reason to resist.
- **Deliver pre-call briefs.** Push a short summary to the rep before they dial: budget, timeline, unit preference, stated objections. The AI arrives as something that saves them ten minutes, not something that replaces them.
- **Attribute wins to the closer.** Site visits sourced by the AI are credited to the representative who converts them. Make the team look effective, not redundant.
- **Instrument follow-up latency.** Track the time between AI qualification and human follow-up as a pilot metric. This turns a suspicion into a number and lets the sponsor manage it directly.
- **Position consistently.** The AI handles the routine eighty percent so the team can concentrate on the high-value twenty percent. This framing should appear in the sales conversation and in the internal rollout, identically.

> **Verdict:** This requires no engineering. It is a deployment design decision, and it should be a condition of every pilot agreement rather than something we discover during the retrospective.

Challenge 8 — Proving ROI Within a Short Pilot Window

Risk Level	Root Cause Category	Engineering Effort	Impact on Runtime Cost
High	Commercial model	Zero engineering	None

The Challenge

Real estate conversion cycles run long. A lead qualified today may visit in two weeks and purchase three months later. A thirty-day pilot therefore ends before any closed revenue is attributable. The developer sees calls made and visits booked, cannot yet see deals closed, and reasonably questions whether the visits were of adequate quality. Renewal conversations become difficult at exactly the moment we have the least evidence.

Why It Matters

- This determines whether technically successful pilots convert into contracts, which makes it a revenue problem rather than a reporting problem.
- Without a comparison baseline, any result we present is arguable — the developer can always attribute the visits to their existing team or to market conditions.
- The remedy must be in place from day one of the pilot; it cannot be constructed retrospectively.

Proposed Solution

Price on the outcome that is attributable inside the pilot window, and build a counterfactual from the first day so that our results are demonstrable rather than assertable.

Key Implementation Points

- **Price per qualified site visit.** A visit occurs inside the pilot window and is unambiguously attributable. A closed deal is neither. The commercial unit should match the measurement window.
- **Run a randomised holdout.** Route twenty percent of pilot leads to the human team only, assigned at random from day zero. At renewal we present a comparison rather than a testimonial. This costs nothing and is the strongest commercial asset obtainable within a pilot.
- **Define visit quality up front.** Agree the qualification criteria with the developer before the pilot starts, so that "were these good visits" is answered by a pre-agreed definition rather than by argument afterwards.
- **Continue tracking past pilot end.** Maintain funnel tracking after the pilot concludes so that delayed conversions are available as evidence at the renewal conversation.
- **Report against their existing cost.** Present cost per qualified visit alongside the developer's current blended cost of achieving the same outcome. Relative numbers persuade; absolute numbers invite scrutiny.

> **Verdict:** Zero engineering cost and the single highest-leverage commercial decision available to us. The holdout in particular is rarely done by competitors and converts our pilot report from a claim into evidence.

11. Summary

Three conclusions follow from the analysis above.

- **The differentiator is not voice quality.** Competitive voice synthesis for Indian languages is now widely available. The teams winning deployments are the ones that solved the unglamorous problems — data hygiene, latency on cellular networks, answer accuracy, clean handoff to humans, and adoption inside the customer organisation.
- **Effectiveness and cost mostly agree.** This was the central concern behind this review, and the finding is favourable. Tiered routing, prompt caching, structured fact lookup and the pre-dial gate each improve quality and reduce cost simultaneously. The genuine trade-offs are narrower than expected and are listed in section 12.
- **Two of the eight challenges require no engineering at all.** Adoption and ROI proof are solved by how we structure the pilot and price the product. They are also two of the most common reasons pilots fail to renew, which makes them disproportionately valuable relative to their cost.

The recommended position is that the first two weeks of work should go to the pre-dial data gate, the compliance layer and structured fact retrieval. These three remove the most common causes of pilot failure. Cost optimisation through tiered routing follows immediately after, because that is where gross margin is determined.

11.1 Consolidated Summary Table

#	Challenge	Core Risk If Unaddressed	Primary Solution	Eng. Effort	Runtime Cost
1	Uncanny valley	Early-call hang-ups cap all downstream performance	Engineered imperfection: fillers, micro-pauses, barge-in, real-network testing	Low	Neutral
2	Latency on Indian networks	Conversations break; product fails in production despite good demos	Itemised latency budget; semantic endpointing, streaming STT, prompt caching, Mumbai hosting	Medium	Reduces
3	Dirty data and CRM chaos	Visible errors blamed on our product; pilot terminated	Deterministic pre-dial data quality gate as a hard blocker	Low	Reduces
4	Tightening regulation	Core positioning invalidated by disclosure mandate	Disclosure-first design, keyword opt-out, consent ledger, DPDP alignment	Low	Neutral
5	First thirty seconds	Wrong project answers lose leads permanently and damage developer brand	Structured facts by deterministic lookup; retrieval restricted to prose	Medium	Reduces
6	Multi-turn complexity	Agent collapses on the first unexpected turn	Digression stack, confidence gating, golden set from fifty real calls	Medium	Slight increase
7	Adoption gap	Pilot fails on organisational resistance while technology performs correctly	Pilot scoped to unwanted lead pool; pre-call briefs; attribution to closers	None	None
8	ROI within pilot window	Technically successful pilots do not convert to contracts	Price per qualified visit; twenty percent randomised holdout from day zero	None	None

12. Cost–Efficiency Position

The brief for this review was that solutions must be effective without being expensive. The table below states explicitly where we should spend and where we should not, so that the trade-off is a documented decision rather than an implicit one.

Spend Here	Rationale	Save Here	Rationale
Endpointing and turn-detection quality	The largest single contributor to perceived latency, and the cheapest place to recover it	Premium voice fidelity beyond natural	Now table stakes; additional fidelity moves us toward the uncanny valley, not away from it
Prompt caching and the small-model tier	Reduces both latency and token cost on every turn	Frontier model on every conversational turn	The majority of turns are deterministic or simple; paying frontier rates for them is pure waste
Structured knowledge base per project	Directly determines answer accuracy, which determines conversion	Larger embedding models and broad retrieval	Adds token cost and latency without improving accuracy on structured project data
Golden-set regression from real calls	The only reliable way to find edge cases before customers do	Full audio evaluation on every iteration	Text-mode regression catches most defects at a fraction of the cost; sample audio instead
India-region hosting and domestic media path	Removes avoidable network delay that no software change can recover	Self-hosted speech infrastructure at current volume	API pricing wins below the concurrency break-even point; revisit when volume justifies it

> **On self-hosting speech models:** The break-even point is approximately hourly GPU cost divided by (API per-minute rate x 60 x utilisation rate). Below that concurrency, the API is cheaper. This calculation should be run against confirmed vendor rates before any infrastructure investment is approved.

12.1 Implementation Sequence

Ordered by return on effort rather than by technical dependency, so that the earliest work removes the largest risks.

Order	Workstream	Addresses	Rationale for Position
1	Pre-dial data quality gate	Challenge 3	Cheapest to build, prevents the most visible and most common pilot failure
2	Disclosure, opt-out and consent ledger	Challenge 4	Approximately one day of work; removes regulatory exposure entirely
3	Structured fact retrieval and KB onboarding template	Challenge 5	The accuracy moat; also reduces runtime cost relative to broad retrieval
4	Three-tier turn router and prompt caching	Challenges 2, 6	Where gross margin is determined; requires stable flows to tier against
5	Latency line items and filler injection	Challenges 1, 2	Depends on the router being in place to know where fillers are needed

Order	Workstream	Addresses	Rationale for Position
6	Digression stack and confidence gating	Challenge 6	Refinement layer; needs the core dialogue flow stable first
7	Golden-set regression harness	Challenges 5, 6	Locks in everything above and protects it against future regression
—	Pilot design: lead pool scoping, holdout, pricing unit	Challenges 7, 8	No engineering dependency; must be agreed before the first pilot begins

Items one through three represent approximately two weeks of work and remove the three most common causes of pilot failure. Items four and five are where our gross margin is established. Items seven and the pilot design items should be settled before the first customer engagement rather than during it.

13. Risks, Assumptions and Open Questions

The following are stated explicitly so that they can be tested rather than carried forward as unexamined premises.

13.1 Assumptions Requiring Validation

- **Tier distribution.** The share of turns handled at each tier is a hypothesis based on the structure of a presales script, not a measurement. It must be validated against a real call sample before it is used in financial modelling.
- **Vendor rates.** All cost reasoning in this document is relative rather than absolute. Current telephony, STT, TTS and inference rates must be confirmed before any pricing commitment.
- **Sub-500ms target.** This target is derived from the point at which conversation quality degrades. It should be confirmed against measured hang-up rates on our own calls once we have volume.
- **Regulatory direction.** Mandatory AI disclosure is our reading of the direction of policy, not present law. The disclosure-first design is defensible regardless of whether the mandate arrives.

13.2 Residual Risks

Risk	Impact	Mitigation
Frontier-tier turns exceed projected share, raising per-minute cost above the price point	Margin compression; possible repricing	Instrument tier distribution from the first production call; set an alert threshold and tune routing before it becomes commercial
Knowledge base onboarding proves more manual than expected, limiting how fast we can add projects	Constrains scaling regardless of platform capability	Measure onboarding hours per project during the first three; templatise aggressively against actual friction points
Developer CRM data is poor enough that the pre-dial gate rejects most leads	Pilot appears to underperform through no fault of the platform	Report rejection statistics to the customer up front and agree remediation as a pilot precondition
Carrier-side spam flagging degrades connect rates independent of call quality	Reduced reach; harder attribution of poor results	Pacing controls, number rotation, and monitoring of connect rate by number as a standing metric
TTS splice seams are audible enough to undermine the naturalness gains	Partially negates the Challenge 1 work	Splice only at sentence boundaries with matched voice settings; validate on cellular audio before rollout

13.3 Open Questions for Discussion

- Is the twenty percent holdout commercially acceptable to a pilot customer, or does it need to be framed as a quality-assurance control rather than an experiment?
- Do we commit to disclosure-first now, or wait for the regulatory position to settle? The recommendation in this document is to commit now, but it is a positioning decision rather than a purely technical one.
- What is the acceptable ceiling on cost per qualified site visit, and what does that imply for the maximum tolerable frontier-tier share?
- Should knowledge base onboarding be a chargeable professional-services line item, or absorbed as customer acquisition cost?

13.4 Recommendation

Approve workstreams one through three for immediate development, and adopt the pilot design decisions under Challenges 7 and 8 as standard terms for every engagement. Defer the self-hosting question until the concurrency

break-even calculation can be run against confirmed vendor rates. Instrument tier distribution from the first production call so that the central cost assumption in this document is validated with real data rather than carried as an estimate.

14. Glossary

Term	Definition
Barge-in	The ability of a caller to interrupt the agent mid-sentence and have playback stop immediately
CP (Channel Partner)	A broker or agency that sources buyer leads on behalf of a developer
DLT	Distributed Ledger Technology registry mandated by TRAI for registering commercial communication templates and sender identities
DND	Do Not Disturb registry; numbers registered on it may not receive commercial calls
DPDP	Digital Personal Data Protection Act — India's data protection framework
E.164	The international standard format for telephone numbers
Endpointing	Detecting when a speaker has finished their turn so the system can begin responding
Golden set	A fixed collection of real conversation transcripts used as regression tests on every system change
Holdout	A randomly selected group excluded from treatment, used as a comparison baseline
LLM	Large Language Model — generates the agent's conversational responses
RAG	Retrieval-Augmented Generation — retrieving relevant documents and supplying them to the model as context
RERA	Real Estate Regulatory Authority; projects carry a mandatory RERA registration number
STT	Speech-to-Text; converts caller audio into text
Time-to-first-token	Delay between sending a request to a model and receiving the first output token
TRAI	Telecom Regulatory Authority of India
TTS	Text-to-Speech; converts the agent's text response into audio
Warm transfer	Handing a live call to a human agent with the conversation context already summarised
8kHz codec	The narrowband audio compression used on standard telephony, materially lower quality than the wideband audio used in browser demos